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A dual function of the IDA peptide in regulating cell separation and modulating plant immunity at the molecular level

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Abstract

The abscission of floral organs and emergence of lateral roots in Arabidopsis is regulated by the peptide ligand INFLORESCENCE DEFICIENT IN ABSCISSION (IDA) and the receptor protein kinases HAESA (HAE) and HAESA-LIKE 2 (HSL2). During these cell separation processes, the plant induces defense-associated genes to protect against pathogen invasion. However, the molecular coordination between abscission and immunity has not been thoroughly explored. Here we show that IDA induces a receptor-dependent release of cytosolic calcium ions (Ca^{2+}) and apoplastic production of reactive oxygen species, which are signatures of early defense responses. In addition, we find that IDA promotes late defense responses by the transcriptional upregulation of genes known to be involved in immunity. When comparing the IDA induced early immune responses to known immune responses, such as those elicited by flagellin22 treatment, we observe both similarities and differences. We propose a molecular mechanism by which IDA promotes signatures of an immune response in cells destined for separation to guard them from pathogen attack.

eLife assessment

This **valuable** study provides insights into the IDA peptide with dual functions in development and immunity. The approach used is **solid** and helps to define the role of IDA in a two-step process, cell separation followed by activation of innate defenses. The main limitation of the study is the lack of direct evidence linking signaling by IDA and its HAE receptors to immunity. As such the work remains descriptive but it will nevertheless be of interest to a wide range of plant cell biologists.

Introduction

All multicellular organisms require tightly regulated cell-to-cell communication during development and adaptive responses. In addition to hormones, plants use small-secreted peptide ligands to regulate these highly coordinated events of growth, development and responses to abiotic and biotic stress (Matsubayashi, 2011; Olsson et al., 2018). In plants, organ separation or abscission, involves cell separation between specialized abscission zone (AZ) cells and enables the removal of unwanted or diseased organs in response to endogenous developmental signals or environmental stimuli. *Arabidopsis thaliana* (Arabidopsis) abscise floral organs (stamen, petals and sepals) after pollination, and this system has been used as a model to study the regulation of cell separation and abscission (Bleecker & Patterson, 1997). Precise cell separation also occurs during the emergence of new lateral root (LR) primordia, root cap sloughing, formation of stomata and radicle emergence during germination (Roberts et al., 2000). In Arabidopsis, the INFLORESCENCE DEFICIENT IN ABSCISSION (IDA) peptide mediated signaling system ensures the correct spatial and temporal abscission of sepals, petals, and stamens. The IDA protein shares a conserved C-terminal domain with the other 7 IDA-LIKE (IDL) members (Butenko et al., 2003; Vie et al., 2015) which is processed to a functional 12 amino acid hydroxylated peptide (Butenko et al., 2014). The abscission process is regulated by the production and release of IDA, which binds the genetically redundant plasma membrane (PM) localized receptor kinases (RKs), HAESA (HAE) and HAESA-LIKE 2 (HSL2). IDA binding to HAE and HSL2 promotes receptor association with members of the co-receptor SOMATIC EMBRYOGENESIS RECEPTOR KINASE (SERK) family and further downstream signaling leading to cell separation events. (Cho et al., 2008; Meng et al., 2016; Santiago et al., 2016; Stenvik et al., 2008). A deficiency in the IDA signaling pathway prevents the expression of genes encoding secreted cell wall remodeling and hydrolase enzymes thus hindering floral organs to abscise (Butenko et al., 2003; Cho et al., 2008; Kumpf et al., 2013; Meng et al., 2016; Niederhuth et al., 2013; Stenvik et al., 2008; Kumpf et al., 2013). In addition, IDL1 signals through HSL2 to regulate cell separation during root cap sloughing (Shi et al., 2018).

Plant cell walls act as physical barriers against pathogenic invaders. The cell-wall processing and remodeling that occurs during abscission leads to exposure of previously protected tissue. This provides an entry point for phytopathogens thereby increasing the need for an effective defense system in these cells (Agustí et al., 2009). Indeed, it has been shown that AZ cells undergoing cell separation express defense genes and it has been proposed that they function in protecting the AZ from infection after abscission has occurred (Cai & Lashbrook, 2008; Niederhuth et al., 2013). *mIDA* elicits the production of Reactive Oxygen Species (ROS) in *Nicotiana benthamiana* (*N.benthamiana*) leaves transiently expressing the HAE or HSL2 receptor (Butenko et al., 2014). As a consequence of defense responses, ROS occurs in the apoplast and is involved in regulating the cell wall structure by producing cross-linking of cell wall components in the form of hydrogen peroxide (H_2O_2) or act as cell wall loosening agents in the form of hydroxyl radical (.OH)radicals (Kärkönen & Kuchitsu, 2015). ROS may also serve as a direct defense response by their highly reactive and toxic properties (Kärkönen & Kuchitsu, 2015). In addition, ROS may act as a secondary signaling molecule enhancing intracellular defense responses as well as contributing to the induction of defense related genes (Reviewed in (Castro et al., 2021)). In Arabidopsis, apoplastic ROS is mainly produced by a family of NADPH oxidases located in the plasma membrane known as RESPIRATORY BURST OXIDASE PROTEINs (RBOHs) (Torres & Dangl, 2005). The production of ROS is often found closely linked to an increase in the concentration of cytosolic Ca^{2+} ($[Ca^{2+}]_{cyt}$), where the interplay between these signaling molecules is observed in immunity across kingdoms (Kadota et al., 2015; Steinhorst & Kudla, 2013). Interestingly, ROS and $[Ca^{2+}]_{cyt}$ are known signaling molecules acting downstream of several peptide ligand-receptor pairs. This includes the endogenous PAMPs ELICITOR PEPTIDE(PEP)1, PEP2 and PEP3 which function as elicitors of systemic responses against pathogen attack and herbivory (Huffaker et al., 2006; Ma et al., 2012; Qi et al., 2010), as well as the pathogen derived peptide ligands flagellin22 (flg22) and Ef-Tu which bind and activate defense related RKs in the PM of the plant cell (Felix et al., 1999; Gómez-Gómez et al., 1999; Ranf et al., 2011).

The interplay between ROS and $[Ca^{2+}]_{cyt}$ has been well studied in the flg22 induced signaling system. The flg22 peptide interacts with the extracellular domain of the FLAGELLIN SENSING2 (FLS2) receptor kinase, which rapidly leads to an increase in $[Ca^{2+}]_{cyt}$ (Ranf et al., 2011). In a resting state, the co-receptor BRASSINOSTEROID INSENSITIVE1-ASSOCIATED RECEPTOR KINASE1/SERK3) (BAK1) and the cytoplasmic kinase BOTRYTIS-INDUCED KINASE1 (BIK1) form a complex with FLS2. Upon binding of flg22 to the FLS2 receptor, BIK1 is released from the receptor complex and activates intracellular signaling molecules through phosphorylation, among other, the PM localized RBOHD (Kadota et al., 2014; Li et al., 2014). flg22 induced apoplastic ROS production binds to PM localized Ca^{2+} channels on neighboring cells, activating Ca^{2+} influx into the cytosol. The cytosolic Ca^{2+} may bind to CALCIUM DEPENDENT PROTEIN KINASE5 (CDPKs) regulating RBOHD activity through phosphorylation. The Ca^{2+} dependent phosphorylation of RBOH enhances ROS production, which in turn leads to a propagation of the ROS/Ca^{2+} signal through the plant tissue (Dubiella et al., 2013)

A rapid increase in $[Ca^{2+}]_{cyt}$ is not only observed in peptide-ligand induced signaling. Similar $[Ca^{2+}]_{cyt}$ changes are observed as responses to developmental signals, such as extracellular auxin and environmental signals such as high salt conditions and drought(reviewed in (Kudla et al., 2018)). Also, $[Ca^{2+}]_{cyt}$ signals are observed in single cells during pollen tube and root hair growth (Monshausen et al., 2008; Sanders et al., 1999). How cells decipher the $[Ca^{2+}]_{cyt}$ changes into a specific cellular response is still largely unknown, however the cell contains a large toolkit of Ca^{2+} sensor proteins that may affect cellular function through changes in protein phosphorylation and gene expression. Each Ca^{2+} binding protein has specific affinities for Ca^{2+} allowing them to respond to different Ca^{2+} concentrations providing a specific cellular response (Geiger et al., 2010; Scherzer et al., 2012).

Due to our previous studies showing IDA induced apoplastic ROS production (Butenko et al., 2014), and the known link between ROS and $[Ca^{2+}]_{cyt}$ response observed for other ligands such as flg22 (Dubiella et al., 2013), we sought to investigate if IDA induces a release of $[Ca^{2+}]_{cyt}$. Here we report that IDA is able to induce an intracellular release of Ca^{2+} in Arabidopsis root tips as well as in the AZ. Therefore, we aimed to understand how the observed production of ROS and Ca^{2+} is involved in the IDA-HAE/HSL2 signaling pathway, either by being a part of the developmental process of abscission or as a part of a suggested involvement of IDA in regulating plant defense (Patharkar et al., 2017). We show that a range of biotic and abiotic signals can induce *IDA* expression and we demonstrate that the IDA signal modulates responses of plant immune signaling. We propose that, in addition to regulating cell separation, IDA plays a role in enhancing a defense response in tissues undergoing cell separation thus providing the essential need for enhanced defense responses in tissue during the cell separation event.

Results

IDA induces apoplastic production of ROS and elevation in cytosolic calcium-ion concentration

We have previously shown that a 12 amino acids functional IDA peptide, hereby referred to as mIDA (Sup Table 1), elicits the production of a ROS burst in *N.benthamiana* leaves transiently expressing *HAE* or *HSL2* (Butenko et al., 2014). To investigate whether mIDA could elicit a ROS burst in Arabidopsis, a luminol-dependent ROS assay on *hae hsl2* mutant rosette leaves expressing the full length HAE receptor under a constitutive promoter (*35S:HAE-YFP*) was performed. *hae hsl2 35S:HAE-YFP* leaf discs treated with mIDA emitted extracellular ROS, whereas no response was observed in wild-type (WT) leaves most likely due to the absence of *HAE* expression (Sup Fig. 1). Since rapid production of ROS is often linked to an elevation in $[Ca^{2+}]_{cyt}$ (Steinhorst & Kudla, 2013), we aimed to investigate if mIDA could induce a $[Ca^{2+}]_{cyt}$ response. We imaged Ca^{2+} in 10 days-old roots expressing the cytosolic localized fluorescent Ca^{2+} sensor R-GECO1 (Keinath et al., 2015). As a positive control for $[Ca^{2+}]_{cyt}$ release we added 1mM extracellular ATP (eATP), which leads to a $[Ca^{2+}]_{cyt}$ elevation in the root tip within a minute after application (Fig. 1a) (Breiden et al., 2021). After application of 1μM mIDA, we detected an increase in $[Ca^{2+}]_{cyt}$ (Fig. 1a and Movie 1). R-GECO1 fluorescence intensities normalized to background intensities (ΔF/F) were measured from a region of interest (ROI) covering the meristematic and elongation zone of the root and revealed that the response starts 4-5 minutes after application of the mIDA peptide and lasts for 7-8 minutes (Fig. 1a). The signal initiated in the root meristematic zone from where it spread toward the elongation zone and root tip, a second wave was observed in the meristematic zone continuing with Ca^{2+} spikes. The signal amplitude was at a maximum within the elongation zone and decreased as the signal spread (Fig. 1a).

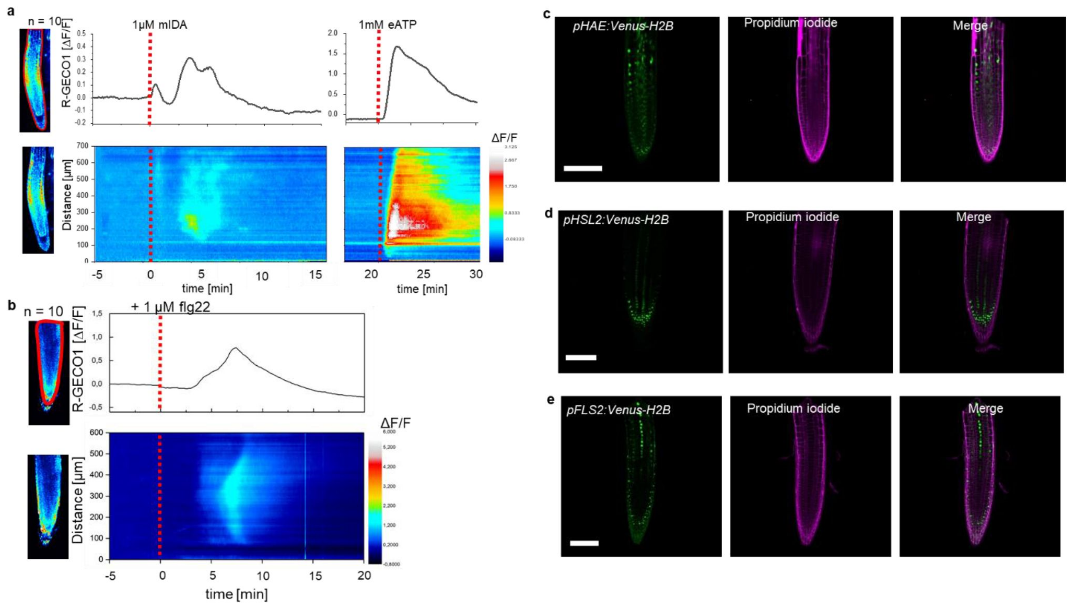

Fig. 1:

mIDA-induced $[Ca^{2+}]_{cyt}$ release in Arabidopsis roots correlates with *pHAE* and *pHSL2* activity

a, Normalized R-GECO1 fluorescence intensities (ΔF/F) were measured from regions of interest (ROI) (upper panel, outlined in red) in the meristematic and elongation zone of the root. Fluorescence intensities (ΔF/F) over time of the whole root represented in a heat map (lower panel). Shown are cytosolic calcium concentration ($[Ca^{2+}]_{cyt}$) dynamics in the ROI in response to 1 mM mIDA over time. (see also Movie 1). Red lines at 5 minutes (min) indicates application of mIDA peptide or application of eATP at 22 min. Representative response from 10 roots. The increase in $[Ca^{2+}]_{cyt}$ response propagates through the roots as two waves. **b,** For comparison; Normalized R-GECO1 fluorescence intensities (ΔF/F) measured from regions of interest (ROI) (outlined in red, upper panel) in response to 1μM flg22 over time. Fluorescence intensities (ΔF/F) over time of the whole root represented in a heat map (lower panel). Red line at 0 min (min) indicates application of flg22 peptide. Representative response from 10 roots. The increase in $[Ca^{2+}]_{cyt}$ response propagates through the roots as a single wave seen as normalized R-GECO1 fluorescence intensities (ΔF/F) shown as a heat map (see also movie 2). **c,d,e,** Expression of the receptors c, *pHAE:Venus-H2B* d, *pHSL2:Venus-H2B* and e, *pFLS2:Venus-H2B* in 7 days-old roots. Representative pictures of n = 8. Scale bar = 50 µm, single plane image, magenta = propidium iodide stain.

The $[Ca^{2+}]_{cyt}$ response in roots to the bacterial elicitor flg22 has previously been studied using the R-GECO1 sensor (Keinath et al., 2015). To better understand the specificity of the mIDA induced Ca^{2+} response, we aimed to compare the Ca^{2+} dynamics in mIDA treated roots to those treated with flg22. We observed striking differences in the onset and distribution of the Ca^{2+} signals. Analysis of roots treated with 1 µM flg22 showed that the Ca^{2+} signal initiated in the root elongation zone from where it spread toward the meristematic zone as a single wave and that the signal amplitude was at a maximum within the elongation zone and decreased as the signal spread (Fig. 1b, Movie 2). These observations indicate differences in tissue specificity of Ca^{2+} responses between mIDA and flg22, which we hypothesize depend on the cellular distribution of their cognate receptors. Indeed, when investigating the promoter activity of the *HAE*, *HSL2* and *FLS2* receptors by cloning the promoters fused to the nuclear localized YFP-derived fluorophore Venus protein (*pHAE:Venus-H2B*, *pHSL2:Venus-H2B* and *pFLS2:Venus-H2B*) we observed a different pattern of fluorescent nuclei in the roots (Fig. 1c,d,e). *pHAE:Venus-H2B* lines had fluorescent expression in the epidermis and stele of the elongation zone (Fig. 1c) and *pHSL2:Venus-H2B* lines in the lateral root cap, root tip and root meristem (Fig. 1d); while fluorescent nuclei were observed in the stele of the elongation zone in *pFLS2:Venus-H2B lines* (Fig. 1e). The different patterns of fluorescent nuclei observed for the three different receptor constructs could indeed explain the differences in the Ca^{2+} signatures triggered by mIDA and flg22. However, the receptors promoter activity show a broader expression pattern compared to the respective ligand induced $[Ca^{2+}]_{cyt}$ responses, indicating additional regulating factors for the observed $[Ca^{2+}]_{cyt}$ response.

The flg22 induced $[Ca^{2+}]_{cyt}$ response is completely abolished in the *fls2* mutant (Ranf et al., 2011). To investigate the dependency of the mIDA-induced response on HAE and HSL2, a cytosolic localized Aequorin-based luminescence Ca^{2+} sensor (Aeq) was introduced into the *hae hsl2* mutant background (Knight et al., 1991). The Aeq sensor was chosen due to the well-established use of this sensor in mutant studies (Ranf et al., 2011) and due to problems with transgene silencing of the R-GECO1 reporter in *hae hsl2* (not shown). The mIDA triggered increase in $[Ca^{2+}]_{cyt}$ was completely abolished in *hae hsl2 Aeq* seedlings compared to WT *Aeq* expressing seedlings, indicating that the response observed is receptors dependent (Sup Fig. 2a). Removing the last Asparagine of the mIDA peptide (IDAΔN69; Sup Table 1) renders it inactive (Butenko et al., 2014). Seedlings treated with 1 µM IDAΔN69 did not show any increase in $[Ca^{2+}]_{cyt}$ (Sup Fig. 2a). To investigate if the $[Ca^{2+}]_{cyt}$ increase triggered by mIDA was dependent on extracellular Ca^{2+}, we pre-treated seedling with $LaCl_3$ and Ethylene glycol-bis (2-aminoethylether)-N, N, N',N'-tetraacetic acid (EGTA). $LaCl_3$ and EGTA are inhibitors that block plasma membrane localized cation channels and chelate Ca^{2+} in the extracellular space, respectively (Knight et al., 1997). The mIDA induced response was abolished in *Aeq*-seedlings pre-incubated in 2 mM EGTA or 1 mM $LaCL_3$ (Sup Fig. 2b), indicating that the mIDA dependent $[Ca^{2+}]_{cyt}$ response depends on Ca^{2+} from the extracellular space.

Next we set out to investigate whether mIDA would trigger an increase in $[Ca^{2+}]_{cyt}$ in floral AZ cells. We utilized the Aeq expressing line and monitored $[Ca^{2+}]_{cyt}$ changes in flowers at different developmental stages (see Sup Fig. 3a for developmental stages [floral positions]) to 1 µM mIDA. Interestingly, only flowers at a the stage where there is an initial weakening of the AZ cell walls showed an increase in $[Ca^{2+}]_{cyt}$ (Sup Fig. 3). Flowers treated with mIDA prior to cell wall loosening showed no increase in luminescence (Sup Fig. 3), indicating that the mIDA triggered Ca^{2+} release in flowers correlates with the onset of the abscission process and the increase in *HAE* and *HSL2* expression at the AZ (Cai & Lashbrook, 2008; Patharkar & Walker, 2015). Using plants expressing the R-GECO1 sensor we performed a detailed investigation of the mIDA induced $[Ca^{2+}]_{cyt}$ response in flowers at position 6 which is the position where initial cell wall loosening occurs. The AZ region was analyzed for signal intensity values and revealed that the Ca^{2+} signal was composed of one wave (Fig. 2, Movie 3). The signal initiated close to the nectaries and spread throughout the AZ and further into the floral receptacle. Flowers treated with 1 µM IDAΔN69 did not show any increase in $[Ca^{2+}]_{cyt}$ in the AZ or receptacle, while a clear Ca^{2+} wave could be detected in the whole AZ, receptacle and proximal pedicel after treatment with eATP (Sup Fig.4, Sup Movie 2). Detailed investigation of *HAE* and *HSL2* promoter activity in flowers at position 6 shows restricted promoter activity to the AZ cells (Fig. 2b). The observed promoter activity correlates with the observed mIDA induced $[Ca^{2+}]_{cyt}$ response.

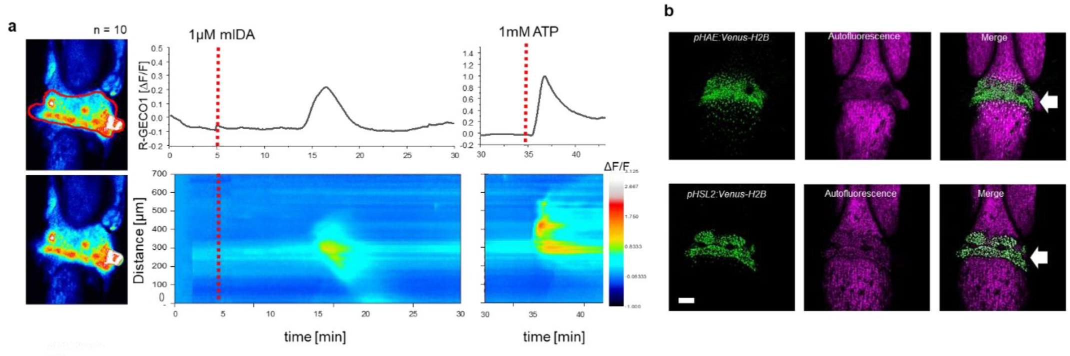

Fig. 2:

mIDA-induced $[Ca^{2+}]_{cyt}$ release in Arabidopsis abscission zones.

a, Normalized R-GECO1 fluorescence intensities (ΔF/F) were measured from regions of interest (ROI) (outlined in red) in floral abscission zone (AZ)s. Shown are cytosolic calcium concentration ($[Ca^{2+}]_{cyt}$) dynamics in the ROI in response to 1 μM mIDA over time (see also Movie 3). Representative response from 10 flowers. Red lines at 5 minutes (min) indicates application of mIDA peptide or application of eATP at 35 min (for AZs), respectively. The increase in $[Ca^{2+}]_{cyt}$ response propagates through the AZ as a single wave. **b,**Expression of *pHAE:Venus-H2B* and *pHSL2:Venus-H2B* in flowers at position 6 (See Sup Fig. 4a for positions) (arrowhead indicates AZ). Representative pictures of n = 8, scale bar = 100 μm, maximum intensity projections of z-stacks. **b** Expression of *pHAE:Venus-H2B* and *pHSL2:Venus-H2B* in flower position 6 (See Sup. Fig.4a for positions) (arrowhead indicates AZ). Representative picture of n = 8. Scale bar = 100 μm, maximum intensity projections of z-stacks.

The ROS producing enzymes, RBOHD and RBOHF, are not involved in the developmental process of abscission

ROS and Ca^{2+} are secondary messengers that can play a role in developmental processes involving cell wall remodeling, but are also important players in plant immunity (Kärkönen & Kuchitsu, 2015). We therefore set out to investigate if the IDA induced production of ROS and increase in $[Ca^{2+}]_{cyt}$ function in the developmental process of abscission or if these signaling molecules form part of IDA modulated plant immunity.

The IDA induced Ca^{2+} response depends on Ca^{2+} from extracellular space (Sup Fig. 2b). Ca^{2+} transport over the plasma membrane is enabled by a variety of Ca^{2+} permeable channels, including the CYCLIC NUCLEOTIDE GATED CHANNELs (CNGCs). Various CNGCs are known to be involved in peptide ligand signaling in plants, including the involvement of CNGC6 and CNGC9 in CLAVATA3/EMBRYO SURROUNDING REGION40 signaling in roots (Breiden et al., 2021), and CNGC17 In phytosulfokine signaling (Ladwig et al., 2015). Using publicly available expression data we identified multiple CNGCs expressed in the AZ of Arabidopsis (Cai & Lashbrook, 2008) (Sup Fig. 5a) and investigated if plants carrying mutations in the *CNGSs* genes expressed in AZs showed a defect in floral organ abscission. We observed no deficiency in floral abscission in the *cngc* mutants (Sup Fig. 5b), in contrast to what is observed in the *hae hsl2* mutant where all floral organs are retained throughout the inflorescence. Ca^{2+} partially activates members of the NOX family of NADPH oxidases (RBOHs), key producers of ROS (Kadota et al., 2015). Two members of this family, *RBOHD* and

RBOHF have high transcriptional levels in the floral AZs (Cai & Lashbrook, 2008) and have been reported to be important in the cell separation event during floral abscission in Arabidopsis (Lee et al., 2018). We therefore set out to investigate if ROS production in AZ cells was dependent on RBOHD and RBOHF and to quantify to which degree these NADPH oxidases were necessary for organ separation. We treated AZs with the ROS indicator, H2DCF-DA, which upon contact with ROS has fluorescent properties, and observed ROS in both WT and to a lower extent in *rbohd rbohdf* flowers. In addition, we observed normal floral abscission in the *rbohd rbohf* double mutant flowers, indicated that neither ROS production in the AZ nor cell separation was dependent on RBOHD and RBOHF (Sup Fig. 5c,d). These results are in stark contrast to what has previously been reported, where cell separation during floral abscission was shown to be dependent on RBOHD and RBOHF (Lee et al., 2018). We used a stress transducer to quantify the force needed to remove petals, the petal breakstrength (pBS) (Stenvik et al., 2008), from the receptacle of gradually older flowers along the inflorescence of WT and *rbohd rbohf* flowers. Measurements showed a significant lower pBS value for the *rbohd rbohf* mutant compared to WT at the developmental stage where cell loosening normally occurs, indicating premature cell wall loosening (Sup Fig. 5e). Furthermore, *rbohd rbohf* petals abscised one position earlier than WT (Sup Fig. 5e). In conclusion, we found no RBOHD and RBOHF dependency for cell separation during floral abscission (Crick et al., 2022). Due to a known function of IDA in inducing ROS production (Sup fig 1), these results points towards a role for IDA in modulating additional responses to the process of cell separation.

IDA is upregulated by biotic and abiotic factors

In Arabidopsis, infection with the pathogen *P.syringae* induces cauline leaf abscission. Interestingly, stress induced cauline leaf abscission was reduced in plants with mutations in components of the IDA-HAE/HSL2 signaling pathway (Patharkar et al., 2017; Patharkar & Walker, 2016). In addition, upregulation of defense genes in floral AZ cells during the abscission process is altered in the *hae hsl2* mutant (Cai & Lashbrook, 2008; Niederhuth et al., 2013), indicating that the IDA-HAE/HSL2 signaling system may be involved in regulating defense responses in cells undergoing cell separation. To further investigate the involvement of IDA in stress responses we investigated if *IDA* is upregulated by biotic and abiotic elicitors. Transgenic plants containing the *β-glucuronidase (GUS*) gene under the control of the *IDA* promoter (*pIDA:GUS*) have previously been reported to exhibit *GUS* expression in cells overlaying newly formed LR primordia (Butenko et al., 2003; Kumpf et al., 2013). To investigate the expression of the *IDA* gene in response to biotic stress *pIDA:GUS* seedlings were exposed to the bacterial elicitor flg22 and fungal chitin. Compared to the control seedlings, the seedlings exposed to flg22 and chitin showed a significant increase in *GUS* expression in LR primordia (Fig. 3a). Also, when comparing untreated root tips to those exposed to flg22 and chitin we observed a GUS signal in the primary root of flg22 and chitin treated roots (Fig. 3b). We used the fluorogenic substrate 4-methylumbelliferyl β-D-glucuronide (4-MUG) which is hydrolysed to the fluorochrome 4-methyl umbelliferone (4-MU) to quantify GUS-activity (Blazquez, 2007). A significant increase in fluorescence was observed for flg22 and chitin treated samples compared to untreated controls, (Fig. 3b). Exposure of *pIDA:GUS* seedlings to IDAΔN69 did not increase GUS activity in the root, supporting the specific increment of *pIDA:GUS* expression to flg22 peptide (Fig. 3). Spatial *pIDA:GUS* expression was also monitored upon abiotic stress. When *pIDA:GUS* seedling were treated with the osmotic agent mannitol or exposed to salt stress by NaCl, both treatments simulating dry soil, a similar increment in GUS signal was observed surrounding the LR primordia (Fig. 3). However, no *pIDA:GUS* expression was detected in the primary root (Fig. 3b). Taken together these results indicate an upregulation of *IDA* in tissue involved in cell separation processes such as the LR primordia and, interestingly, the root cap upon biotic stress. The process of cell separation is a fast and time-restricted response where cells previously protected by outer cell layers are rapidly exposed to the environment. These cells need strong and rapid protection against the environment, and we therefore aimed to investigate if IDA is involved in modulating immune responses in addition to inducing ROS and $[Ca^{2+}]_{cyt}$ responses.

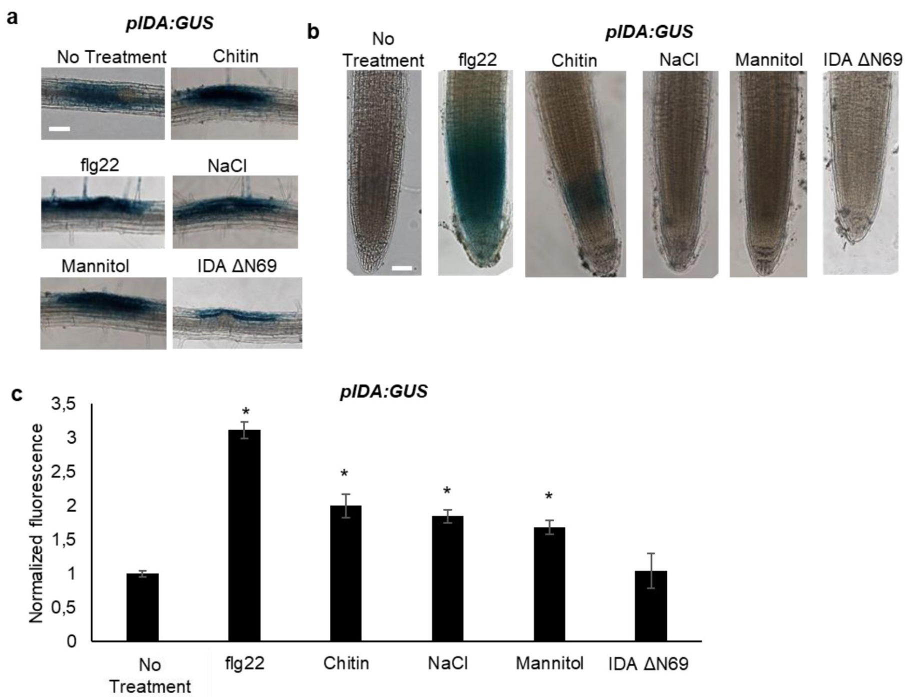

Fig. 3:

IDA is induced by biotic and abiotic stress

Representative pictures of *pIDA:GUS* expression after 12 h treatment with 1 μM flg22, chitin, NaCl, Mannitol and 1 μM IDAΔN69 in **a,** cells surrounding emerging lateral roots, and **b,** in the main root. **c,** Normalized emitted fluorescence of fluorochrome 4-methyl umbelliferone (4-MU) in 7 days-old seedlings after 12 h treatment with 1 μM flg22, chitin, NaCl, Mannitol and 1 μM of the inactive IDA peptide, IDAΔN69. Normalized to 1 on No Treatment sample. Controls were not subjected to any stimuli (No treatment). n = 6, experiment repeated 3 times. * = significantly different from the non-treated (No treatment) sample ($p < 0.05$, student t-test, two tailed). Controls were not subjected to any stimuli (No treatment). **a,b,** Representative picture of n = 10, experiment repeated 3 times, scale bar = 50 μm.

mIDA does not induce production of callose as a long term defense response

We aimed to investigate a possible role of IDA in regulating long-term defense responses in the plant, such as the production of callose. Callose deposition is a resulting hallmark of the late plant immunity responses that together with other events such as stomatal closure and production of ethylene leads to inhibition of pathogen multiplication and containment of disease (Reviewed in (Wang et al., 2021)). Callose deposition is highly increased in response to flg22 (Gómez-Gómez et al., 1999). We treated WT seedlings with 1 μM mIDA and measured the area of callose deposition in the cotyledons, but detected no difference to water treated controls (Fig. 4a,b). However, flg22 treated WT seedlings showed increased callose deposition, which was not observed in flg22 treated *fls2* seedlings (Fig. 4a,b). These results indicate that mIDA, in contrast to flg22, is not involved in promoting deposition of callose as a long-term defense response.

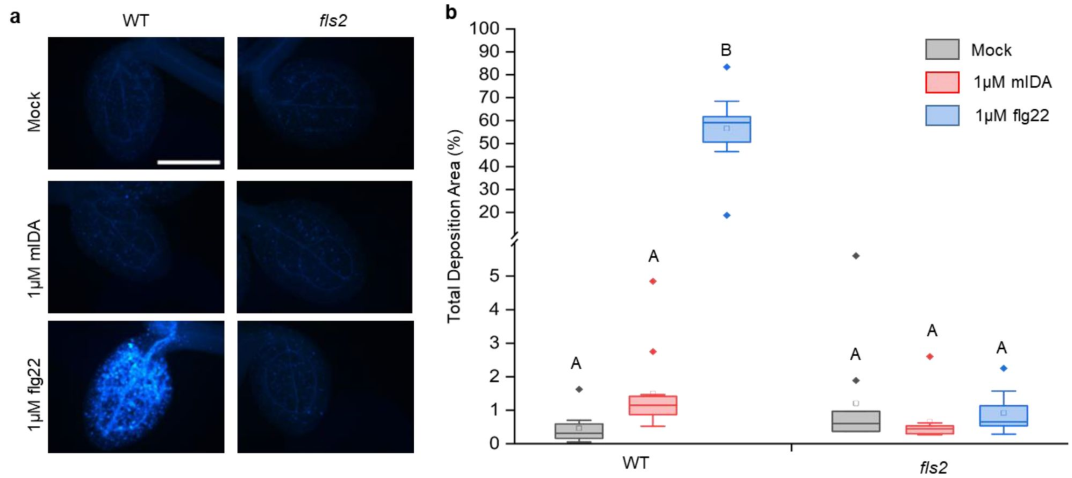

Fig. 4:

Callose deposition is not induced by mIDA

a, No callose deposition could be detected in Col-0 WT and *fls2* mutants treated with water (mock treatment) or 1 µM mIDA. Callose deposition could be observed in cotyledons of eight day old Col-0 WT seedlings treated with 1 µM flg22. No callose deposition could be detected in *fls2* mutant treated with flg22. Representative images of 9-12 seedlings per genotype. **b,** Total callose deposition area for the different genotypes treated with water (mock treatment), 1 µM mIDA or 1 µM flg22. Statistical significant difference at ($p < 1*10^{-6}$). N = 9-12

mIDA triggers expression of defense-associated marker genes associated with innate immunity

A -well-known part of a plants immune response is the enhanced transcription of genes involved in immunity. Transcriptional reprogramming mostly mediated by WRKY transcription factors takes place during microbe-associated molecular pattern (MAMP)-triggered immunity and is essential to mount an appropriate host defense response (Birkenbihl et al., 2017). We selected well-established defense-associated marker genes: *FLG22-INDUCED RECEPTOR-LIKE KINASE1* (*FRK1*), a specific and early immune-responsive gene activated by multiple MAMPs (Asai et al., 2002; He et al., 2006), *MYB51*, a WRKY target gene, previously shown to increase in response to the MAMP flg22 (Frerigmann et al., 2016) resulting in biosynthesis of the secondary metabolite indolic glucosinolates; and an endogenous danger peptide, ELICITOR PEPTIDE 3 (PEP3) (Huffaker et al., 2006). Seven days-old WT seedlings treated with 1µM mIDA were monitored for changes in transcription of the aforementioned genes by RT-qPCR compared to untreated controls at two time points, 1h and 12h after treatment. All genes showed a significant transcriptional elevation 1 h after mIDA treatment (Fig. 5a). This in accordance with previous reports, where the expression of *FRK1* and *PEP3* was monitored in response to bacterial elicitors (He et al., 2006; Huffaker et al., 2006). After 12 h, *FRK1* and *PEP3* transcription in mIDA treated tissue was not significantly different from untreated tissue; while transcription of MYB51 in mIDA treated tissue remained elevated after 12 h but was significantly lower compared to 1 h after mIDA treatment (Fig. 5a). Interestingly, the observed increase in transcription of the defense genes *MYB51* and *PEP3*, was only partially reduced in the *hae hsl2* mutant (Sup Fig. 6). In contrast, *FRK1* showed higher transcriptional levels in the *hae hsl2* mutant compared to WT (Sup Fig. 6). These results indicate that also other receptors than HAE and HSL2 are involved in the mIDA-induced enhancement of the defense genes *FRK1*, *MYB51* and *PEP3*.

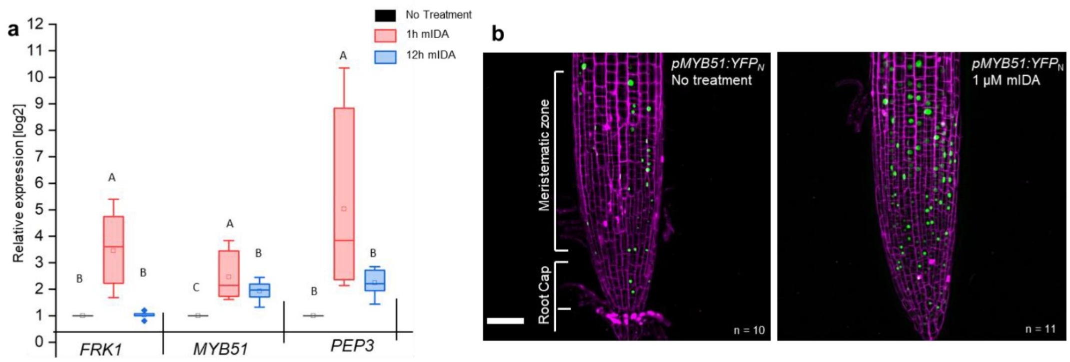

Fig. 5:

mIDA induced expression of defense-associated marker genes

a, Transcripts of *FRK1*, *MYB51*, and *PEP3* in WT Col-0 seedlings exposed to 1 µM mIDA for 1 h (light gray) and to 1 µM mIDA for 12 h (dark gray) compared to untreated tissue (black). RNA levels were measured by RT-qPCR analysis. *ACTIN* was used to normalize mRNA levels. n = 4, three biological replicates Statistical analyses was performed comparing induction times of individual genes using one-way ANOVA and post-hoc Tukey's test ($p < 0.05$). **b,***pMYB51:YFPN* expression is enhanced in roots after 7 h exposure to 1 µM mIDA peptide compared to untreated roots (control), scale bar = 50 µm, maximum intensity projections of z-stacks, magenta = propidium iodide stain.

We then analyzed a plant line expressing nuclear localized YFP from the MYB51 promoter (*pMYB51:YFP^N*) (Poncini et al., 2017) treated with mIDA. Enhanced expression of p*MYB51:YFP* was predominantly detected in the meristematic zone of the root after 7 h of mIDA treatment, compared to a non-treated control (Fig. 5b). Moreover, comparison of mIDA induction with the elicitor-triggered response of *pMYB51:YFP^N* to flg22 showed similar temporal expression in the root (Sup Fig. 7). Together, these data show that IDA can trigger a rapid increase in the expression of key genes involved in immunity.

We next investigated if the mIDA induced transcription of defense related genes was similar to that induced by flg22. Seven days-old Col-0 WT seedlings were treated for 1h with mIDA, flg22 or a combination of mIDA and flg22 to explore potential additive effects. The relative increase in transcription of the genes tested is similar when seedlings are treated with mIDA and flg22 (Fig. 6). However, when treating seedlings with both peptides the relative transcription of the genes of interest increased substantially. To our surprise, the combined enhanced transcription when seedling where co-treated with both flg22 and mIDA exceeds a simple additive effect of the two peptides, suggesting that IDA has a role in enhancing the defense response triggered by flg22. Based on these results we propose a role for IDA in enhancing immune responses in tissue undergoing cell separation.

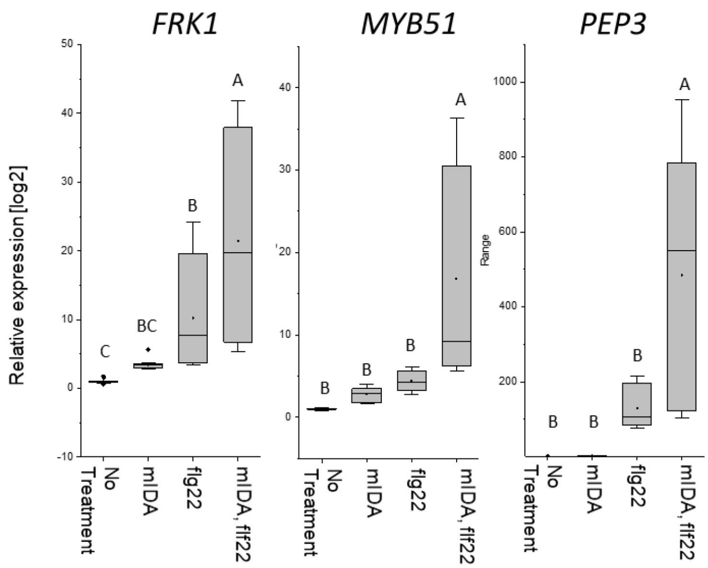

Fig. 6:

mIDA and flg22 co-treatement extremely enhances the expression of defense-associated marker genes

a, RT-qPCR data showing transcription of *FRK1*, *MYB51*, and *PEP3* in Col-0 WT seedlings exposed to 1 μM mIDA, 1 μM flg22 or a combination of 1 μM mIDA and 1 μM flg22 for 1 h compared to untreated seedlings (No treatment control). *ACTIN* was used to normalize mRNA levels. Figure represent 3 biological replicates with 4 technical replicates. Statistical analyses comparing No Treatment to peptide treated samples was performed using one-way ANOVA and post-hoc Tukey's test ($p < 0.05$).

Discussion

Several plant species can abscise infected organs to limit colonization of pathogenic microorganisms thereby adding an additional layer of defense to the innate immune system (Kissoudis et al., 2016; Patharkar et al., 2017). Also, in a developmental context, there is an induction of defense associated genes in AZ cells during cell separation and prior to the formation of the suberized layer that functions as a barrier rendering protection to pathogen attacks (Cai & Lashbrook, 2008; Niederhuth et al., 2013; Roberts et al., 2000; I. Taylor & J. C. Walker, 2018). Interestingly, the induction of defense genes during floral organ abscission is altered in *hae hsl2* plants (Niederhuth et al., 2013). It is largely unknown how molecular components regulating cell separation and how the IDA-HAE/HSL2 signaling pathway, contributes to modulation of plant immune responses. Here we show that the mature IDA peptide, mIDA, is involved in activating known defense responses in Arabidopsis, including the production of ROS, the intracellular release of Ca^{2+} and the transcription of genes known to be involved in defense (Sup Fig. 1, Fig. 1, Fig. 2, Fig. 5). We suggest a role for IDA in ensuring optimal cellular responses in tissue undergoing cell separation, regulating both development and defense.

The intracellular event occurring downstream of IDA is still partially unknown. Here we report the release of cytosolic Ca^{2+} as a new signaling component. Roots and AZs expressing the Ca^{2+} sensor R-GECO1, showed a $[Ca^{2+}]_{cyt}$ in response to mIDA. Moreover, the increased $[Ca^{2+}]_{cyt}$ in response to mIDA in flowers occurred only at stages where cell separation was taking place, linking the observed $[Ca^{2+}]_{cyt}$ response to the temporal point of abscission and the need for an induced local defense response. During floral development the expression of the *HAE* and *HSL2* receptors increase prior to the onset of abscission (Cai & Lashbrook, 2008) and may account for the temporal mIDA induced $[Ca^{2+}]_{cyt}$ response observed.

Based on these results, it is likely that the expression of the receptors is the limiting factor for a cell's ability to respond to the mIDA peptide with a $[Ca^{2+}]_{cyt}$ response. However, promoter activity of the receptors are observed in a broader area of the root than the observed $[Ca^{2+}]_{cyt}$ response. As an example, *pHSL2* is found to have a localized activation in the root tip, and when HSL2 is activated by IDL1 be involved in regulation cell separation during root cap sloughing (Shi et al., 2018). IDL1 and IDA have a high degree of similarity in the protein sequence, and *IDL1* expressed under the *IDA* promoter can fully rescue the abscission phenotype observed in the *ida* mutant (Stenvik et al., 2008). We can assume that IDL1 activates similar signaling components in the root cap as those used by IDA. Despite the expression of HSL2 and the involvement of the receptor in root cap sloughing, no $[Ca^{2+}]_{cyt}$ response is observed in this region when plants are treated with the IDA peptide. The absence of an observed $[Ca^{2+}]_{cyt}$ response in the root tip indicates that $[Ca^{2+}]_{cyt}$ is most likely not involved in the cell separation process during root cap sloughing, or that cells involved in this process only are able to induce a $[Ca^{2+}]_{cyt}$ response at a specific developmental time point. Thus, in addition to receptor availability, other factors must be involved in regulating the $[Ca^{2+}]_{cyt}$ response. It is intriguing that the mIDA cellular output is similar to that of flg22 (Kadota et al., 2015; Kadota et al., 2014) yet shows some distinct differences. The spatial distribution of the Ca^{2+} wave originated by mIDA differs from that of flg22. The *rbohd rbohf* mutant shows ROS production in the AZ, indicating that other NADPH oxidases or cell wall peroxidases expressed in the AZ may be involved in ROS production, in contrast to the importance of RBOHD RBOHF in flg22 signaling (Kadota et al., 2015). This suggests that while IDA and flg22 share many components for their signal transduction, such as the BAK1 co-receptor (Chinchilla et al., 2007; Meng et al., 2016) and MAPK cascade (Asai et al., 2002; Cho et al., 2008), there are other specific components that are likely important to provide differences in signaling. The difference in callose deposition observed in mIDA and flg22 cotyledons supports this.

Interestingly, FLS2 arrangement into nanoscale domains at the PM is dependent on FERONIA (FER). FLS2 becomes more disperse and mobile in *fer* mutants, implying a role for FER in regulating FLS2 activity (Gronnier et al., 2022). Also altering of the cell wall affects FLS2 nanoscale organization (McKenna et al., 2019). Formation of nanoscale domains containing the HAE and HSL2 receptors may be an important signaling step in the IDA-HAE/HSL2 signaling pathway. An intriguing thought is a possible co-localization of HAE and HSL2 with FLS2 in nanoscale domains, making a signaling unit possible to involve multiple different receptors responsible for the signaling outcome observed. In this paper, we show that co-treatment with mIDA and flg22 induces an enhanced transcription of defense related genes. How FLS2, HAE and HSL2 communicate at the plasma membrane of the responsive cells may give the answers to the molecular mechanisms behind the augmented transcriptional regulation observed when co-treatment of IDA and flg22 is performed (Fig. 6). We can investigate possible interaction partners of HAE and HSL2 looking at available data from a sensitized high-throughput interaction assay between extracellular domains (ECDs) of 200 Leucine Rich Repeat (LRR) (Smakowska-Luzan et al., 2018). Interestingly, LRR-RLKs known to play a function in biotic or abiotic stress responses, such as RECEPTOR-LIKE KINASE7 (RLK7), STRUBBELIG-RECEPTOR FAMILY3 , RECEPTOR-LIKE PROTEIN KINASE 1 and FRK1, are found to interact mainly with HSL2, whereas HAE shows interaction with LRRs mainly involved in development (Sup Fig. 8). RLK7 has recently been identified as an additional receptor for CEP4, involved in regulating cell surface immunity and response to nitrogen starvation, indication a role of RLK7 as an additional receptor regulating defense responses in known peptide-receptor complexes (Rzemieniewski et al., 2022). It would be interesting to investigate a possible role of RLK7 in IDA-HAE/HSL2 signaling as a possible signaling partner in the observed IDA-induced early defense responses.

In this paper, we propose a model where the IDA-HAE/HSL2 signaling pathway modulates defense responses in tissues undergoing cell separation. It is beneficial for the plant to ensure an upregulation of defense responses in tissues undergoing cell separation as a precaution against pathogen attack. This tissue is a major entry route for pathogens, and by combining the need of both the IDA peptide and the pathogen molecular pattern, such as flg22, for the induction of a strong defense response, the plant ensures maximum immunity in the most prone cells without spending energy inducing this in all cells (Fig. 7). Such a molecular mechanism, where the presence of mIDA in tissue exposed to stress leads to maximum activation of the immune responses, will ensure maximal protection of infected cells during cell separation, cells which are major potential entry routes for invading pathogens.

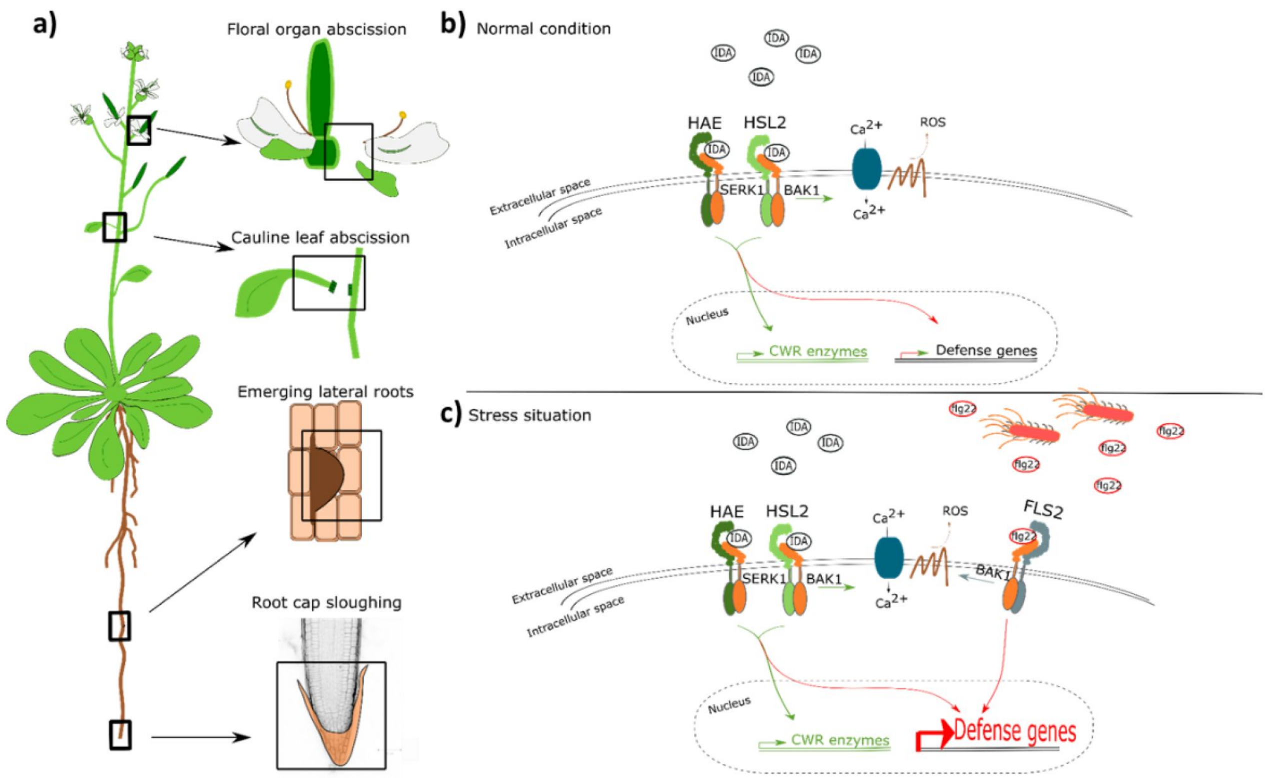

Fig. 7:

IDA regulates cell separation processes, but is also involved in major transcription of defense genes upon pathogen attack.

a, IDA and the IDL peptides control cell separation processes during plant development and in response to abiotic and biotic stress (Butenko et al. 2003; Patharkar and Walker 2016; Patharkar et al. 2017; Shi et al. 2018). Tissue undergoing cell separation includes floral organ abscission, cauline leaf abscission, emerging of lateral roots and root cap sloughing. **b,** During normal conditions, IDA control floral organ abscission and emergence of lateral roots by relaying a signal through receptor complexes including HAE, HSL2, SERK1 and BAK1 to modulate the expression of cell wall remodeling (CWR) genes as well as moderately expression of defense genes. IDA activates a receptor dependent production of ROS and an increase in $[Ca^{2+}]_{cyt}$. **b,** Upon stress, such as pathogen attack, the activation of HAE and HSL2 acts in addition to activation of defense related receptors, such as FLS2, to enhance the expression of defense related genes significantly. This ensure optimal protection of cells undergoing cell separation, which may be major entry routes during a pathogen attack.

Movies

Movie 1: $[Ca^{2+}]_{cyt}$ dynamics in R-GECO1 expressing root tip in response to 1 µM mIDA. 10 days-old roots were treated with 1 µM mIDA and changes in $[Ca^{2+}]_{cyt}$ was recorded over time. Response shown as normalized fluorescence intensities (ΔF/F). Images were recorded with a frame rate of 5 seconds. Movie corresponds to measurements shown in Fig. 1a. Representative response from 10 roots. Time shown as, hh:mm:ss.

Movie 2: $[Ca^{2+}]_{cyt}$ dynamics in R-GECO1 expressing root tip in response to 1 μM flg22 10 days-old roots were treated with 1 μM flg22 and changes in $[Ca^{2+}]_{cyt}$ was recorded over time. Response shown as normalized fluorescence intensities (ΔF/F). Images were recorded with a frame rate of 5 seconds. Movie corresponds to measurements shown in Fig. 1b. Representative response from 10 roots. Time shown as, hh:mm:ss.

Movie 3: $[Ca^{2+}]_{cyt}$ dynamics in R-GECO1 expressing AZ in response to 1 μM mIDA. Flowers at position 6 were treated with 1 μM mIDA and changes in $[Ca^{2+}]_{cyt}$ was recorded over time. Response shown as normalized fluorescence intensities (ΔF/F). Images were recorded with a frame rate of 5 seconds. Movie corresponds to measurements shown in Fig. 2a. Representative response from 10 flowers. Time shown as, hh:mm:ss.

Sup Movie 1: $[Ca^{2+}]_{cyt}$ dynamics in R-GECO1 expressing root tip in response to 1 mM eATP. 1 mM eATP will induce a strong increase in $[Ca^{2+}]_{cyt}$ and was added as a last treatment to all roots acting as a positive control. Response shown as normalized fluorescence intensities (ΔF/F). Images were recorded with a frame rate of 5 seconds. Movie corresponds to measurements shown in Fig. 1a. Representative response from 10 roots. Time shown as, hh:mm:ss.

Sup Movie 2: $[Ca^{2+}]_{cyt}$ dynamics in R-GECO1 expressing AZ in response to to 1 μM of the inactive IDA peptide, $IDA^{\Delta N69}$. Flowers at position 6 were treated with 1 μM $IDA^{\Delta N69}$ and changes in $[Ca^{2+}]_{cyt}$ was recorded over time. Response shown as normalized fluorescence intensities (ΔF/F). Images were recorded with a frame rate of 5 seconds. Movie corresponds to measurements shown in Sup Fig. 3. Representative response from 10 flowers. Time shown as, hh:mm:ss.

Methods

Accession numbers of genes studied in this work

HSL2 At5g65710, IDA At1g68765, FLS2 At5g46330, HAE At4g28490, MYB51 At1g18570, RBOHD At5g47910, RBOHF At1g64060, PEP3 At5g64905, WRKY33 At2g38470, FRK1 At2g19190. Plant lines used in this work: Ecotype Colombia-0 (Col-0) was used as wild type (WT). Mutant line: *hae* (SALK_021905), *hsl2*, (SALK_030520), *fls2* (SALK_062054), *rbohd* (SALK_070610), *rbohf* (SALK_059888), *cngc1* (SAIL_443), *cngc2* (SALK_066908), *cngc4* (SALK_081369), *cngc5* (SALK_149893), *cngc6* (SALK_064702), *cngc9* (SAIL_736), *cngc12* (SALK_093622). SALK lines were provided from Nottingham Arabidopsis Stock Centre (NASC).

Plant lines

The pIDA:GUS, pHSL2:Venus-H2B and pMYB51:YFPN lines have been described previously (Kumpf et al., 2013; Poncini et al., 2017; Shi et al., 2018).The promoters of HAE (1601 bp (Kumpf et al., 2013) and HSL2 (2300 bp (Sto et al., 2015))were available in the pDONRZeo vector (Thermo Fischer Scientific). Sequences corresponding to the FLS2 promoter (988 bp (Robatzek et al., 2006)) were amplified from WT DNA (primers are listed in table 2) and cloned into the pDONRZeo vector (Thermo Fischer Scientific). All promoter constructs were further recombined into the promotor:Venus (YFP)-H2B and Cerulean(CFP)-H2B destination vectors (Somssich et al., 2016) using the Invitrogen Gateway cloning system (Thermo Fischer Scientific). Constructs were transformed into *Agrobacterium tumefaciens* (A.tumefaciens) C58 and the floral dip method (Clough & Bent, 1998) was used to generate transgenic lines. Single-copy homozygous plant lines were selected and used in this study. The CDS of HAE was cloned into the pEarleyGate101 destination vector (Earley et al., 2006) using the Invitrogen Gateway cloning system and transformed into A.tumefaciens C58 and further used to generate the 35S:HAE:YFP lines.

Growth conditions

Plants were grown in long day conditions (8 h dark and 16 h light) at 22 °C. Seeds were surface sterilized and plated out on MS-2 plates, stratified for 24 h at 4 °C and grown on plates for 7 days before transferred to soil.

Peptide sequences

Peptides used in this study were ordered from BIOMATIK. Peptide sequences are listed in Supplementary Table 1.

Primers

Primers for genotyping and generation of constructs were generated using VectorNTI. Gene specific primers for RT-qPCR were generated using Roche Probe Library Primer Design. All primers are listed in Supplementary Table 2.

Histochemical GUS assay

Seven days-old seedlings were pre-incubated for 12 h in liquid MS-2 medium containing stimuli of interest; 1 µM peptide (table 1), 60 mM Mannitol (M4125 – Sigma), 20 µg/mL Chitin (C9752 – Sigma), 50 mM NaCl and then stained for GUS activity following the protocol previously described (Stenvik et al., 2008). Roots were pictured using a Zeiss Axioplan2 microscope with an AxioCam HRc, 20x air objective. The assay was performed on 10 individual roots and the experiment was repeated 3 times.

Fluorescent GUS assay

Seven days-old seedlings were pre-incubated for 12 h in liquid MS-2 medium with or without stimuli of interest; 1 µM peptide (table 1), 60 mM Mannitol (M4125 – Sigma), 20 µg/mL Chitin (C9752 – Sigma), 50 mM NaCl. After treatment, 10 seedlings were incubated in wells containing 1 mL reaction mix described in (Blazquez, 2007) containingcontaining: 10 mM EDTA (pH 8.0), 0,1 % SDS, 50 mM Sodium Phosphate (pH 7.0), 0,1 % Triton X-100, 1 mM 4-MUG (, M9130-Sigma) and incubated at 37 °C for 6 h. Six 100 µl aliquots from each well were transferred to individual wells in a microtiter plate and the reaction was stopped by adding 50 µl of stop reagent (1 M Sodium Carbonate) to each well. Fluorescence was detected by the use of a Wallac 1420 VICTOR2 microplate luminometer (PerkinElmer) using an excitation wavelength of 365 nm and a filter wavelength of 430 nm. Each experiment was repeated 3 times.

Confocal laser microscopy of roots and flowers expressing promoter:Venus/YFP-H2B and promoter: Cerulean-H2B construct

Imaging of 7 days-old roots was performed on a LSM 880 Airyscan confocal microscope equipped with two flanking PMTs and a central 32 array GaAsP detector. Images were acquired with Zeiss Plan-Apochromat 20x/0.8 WD=0.55 M27 objective and excited with laser light of 405 nm, 488 nm and 561 nm. Roots were stained by 1 µM propidium iodide for 10 min and washed in dH_2O before imaging. Imaging of flowers was performed on an Andor Dragonfly spinning disk confocal using an EMCCD iXon Ultra detector. Images were acquired with a 10x plan Apo NA 0,45 dry objective and excited with laser light of 405 nm, 488 nm and 561 nm. Maximum intensity projections of z-stacks were acquired with step size of 1,47 µm. Image processing was performed in FIJI 51. These steps are: background subtraction, gaussian blur/smooth, brightness/contrast. Imaging was performed at the NorMIC Imaging platform.

Calcium imaging using the R-GECO1 sensor

$[Ca^{2+}]_{cyt}$ in roots were detected using WT plants expressing the cytosolic localized single-fluorophore based Ca^{2+} sensor, R-GECO1 (Keinath et al., 2015). Measurements were performed using a confocal laser scanning microscopy Leica TCS SP8 STED 3X using a 20x multi NA 0.75 objective. Images were recorded with a frame rate of 5 seconds at 400 Hz. Seedling mounting was performed as described in (Krebs & Schumacher, 2013). The plant tissues were incubated overnight in half strength MS, 21°C and continuous light conditions before the day of imaging. $[Ca^{2+}]_{cyt}$ in the abscission zone were detected using WT plants expressing R-GECO1 (Keinath et al., 2015). Abscission zones of position 7 were used. Mounting of the abscission zones were performed using the same device as for seedlings (Krebs & Schumacher, 2013). The abscission zones were incubated in half strength MS medium for 1 h before imaging. Measurements were performed with a Zeiss LSM880 Airyscan using a Plan-Apochromat 10× air NA 0.30 objective. Images were recorded with a frame rate of 10 seconds. R-GECO1 was excited with a white light laser at 561 nm and its emission was detected at 590 nm to 670 nm using a HyD detector. Laser power and gain settings were chosen for each experiment to maintain comparable intensity values. For mIDA and flg22 two-fold concentrations were prepared in half strength MS. mIDA or flg22 were added in a 1:1 volume ratio to the imaging chamber (final concentration 1 µM). ATP was prepared in a 100-fold concentration in half strength MS and added as a last treatment in a 1:100 volume ratio (final concentration 1 mM) to the imaging chamber as a positive control for activity of the R-GECO1 sensor (Movie 3). Image processing was performed in FIJI. These steps are: background subtraction, gaussian blur, MultiStackReg v1.45 (http://bradbusse.net/sciencedownloads.html), 32-bit conversion, threshold. Royal was used as a look up table. Fluorescence intensities of indicated ROIs were obtained from the 32-bit images (Krebs & Schumacher, 2013). Normalization was done using the following formula $\Delta F/F = (F-F0)/F0$ where F0 represents the mean of at least 1 min measurement without any treatment. R-GECO1 measurements were performed at the Center for Advanced imaging (CAi) at HHU and at NorMIC Imaging platform at the University of Oslo.

Calcium measurements using the Aequorin (pMAQ2) sensor

$[Ca^{2+}]_{cyt}$ in seedlings and flowers were detected using WT plants expressing p35S-apoaequorin (pMAQ2) located to cytosol (Aeq) (Knight et al., 1991; Ranf et al., 2011). Aequorin luminescence was measured as previously described (Ranf et al., 2012). Emitted light was detected by the use of a Wallac 1420 VICTOR2 microplate luminometer (PerkinElmer). Differences in Aeq expression levels due to seedling size and expression of sensor were corrected by using luminescence at specific time point (L)/Max Luminescence (Lmax). Lmax was measured after peptide treatment by individually adding 100 µL 2 M $CaCl_2$ to each well and measuring luminescence constantly for 180 seconds (Ranf et al., 2012). 2 M $CaCl_2$ disrupts the cells and releases the Aeq sensor into the solution where it will react with Ca^{2+} and release the total possible response in the sample (Lmax) in form of a luminescent peak. A final concentration of 1 µM mIDA was added to each wells at the start of measurements. For inhibitor treatements, Aeq-seedlings were incubated in in 2mM EGTA (Sigma-Aldrich) or 1mM $LaCL_3$ (Sigma-Aldrich) O/N before measurements. For seedlings 3 independent experiments were performed with 12 replications in each experiment. For flowers 3 independent experiments were performed with 4-6 replications in each experiment.

Measurements of reactive oxygen species (ROS)

ROS production was monitored by the use of a luminol-dependent assay as previously described (Butenko et al., 2014) using a Wallac 1420 VICTOR2 microplate luminometer (PerkinElmer). Arabidopsis leaves expressing *35S:HAE:YFP* were cut into leaf discs and incubated in water overnight before measurements. A final concentration of 1 µM mIDA was added to each well at the start of measurements. All measurements were performed on 6 leaf discs and each experiment was repeated 3 times.

ROS stain (H2DCF-DA)

Flowers at position 6 were gently incubated in staining solution (25 µM (2',7'-dichlorodihydrofluorescein diacetate) (H2DCF-DA) (Sigma-Aldrich, D6883), 50 mM KCL, 10 mM MES) for 10 min and further washed 3 times in wash solution (50 mM KCL, 10 mM MES). For the *hae hsl2* mutant the floral organs were forcibly removed immediately before imaging. Imaging was done using a Dragonfly Airy scan spinning disk confocal microscope, excited by a 488 nm laser. A total of 9 flowers per genotype were imaged. The experiment was repeated 2 independent times.

Callose deposition staining and quantification

Callose deposition experiments were conducted as per (Luna et al., 2011) with modifications. Seeds from WT and *fls2* were vapor-phase sterilized for 4 h. Seeds were transferred to 12-well plates containing 1 mL of filter-sterilized MS medium and 0. 5 % MES with a final pH of 5.7, and stratified in the dark at 4 °C for 2 days. Plates were consecutively transferred to a growth chamber with 16 h of light (150µE m-2 s-1)/8h darkness cycle at 22°C. On the seventh day, growth medium was refreshed under sterile conditions. On the eight day the treatments were applied: addition of 10 µL of water (mock); 10 µL of 100 µM flg22; or 10 µL of 100 µM mIDA (final peptide concentrations of 1 µM). After 24 h the growth medium was replaced with 9 5% ethanol and incubated overnight. Ethanol was replaced with an 8 M NaOH softening solution and incubated for 2 h. Seedlings were washed 3 times in water and immediately transferred to the aniline blue staining solution (100 mM KPO4-KOH, pH 11; 0.1 % aniline blue). After 2 h of staining, the seedlings were mounted on 50 % glycerol and imaged under a ZEISS epifluorescence microscope with UV filter (BP 365/12 nm; FT 395 nm; LP 397 nm). Image analysis was performed on FIJI (Schindelin et al., 2012). In brief, cotyledons were cropped and measured the leaf area. Images were thresholded to remove autofluorescence and area of depositions measured to calculate the ratio of callose deposition area per cotyledon area unit. Between 9-12 seedlings were analyzed per genotype x treatment combination.

Real time quantitative PCR (RT-qPCR)

Seven days-old Arabidopsis seedlings (WT, *hae hsl2*) grown vertically on ½ sucrose MS-2 plates were transferred to liquid ½ MS-2 medium (non-treated) and liquid ½ MS-2 medium containing 1 μM peptide and incubated in growth chambers for 1 h or 12 h. Seedlings were flash-frozen in liquid nitrogen before total RNA was extracted using SpectrumTM Plant Total RNA Kit (SIGMA Aldrich). cDNA synthesis was performed as previously described (Grini et al., 2009). RT-qPCR was performed according to protocols provided by the manufacturer using FastStart Essential DNA Green Master (Roche) and LightCycler96 (Roche) instrument. ACTIN2 was used to normalize mRNA levels as described in (Grini et al., 2009). Three biological replicates and 4 technical replicates including standard curves were performed for each sample.

Petal Break strength (pBS)

The force required to remove a petal at a given position on the inflorescence was measured in gram equivalents using a load transducer as previously described (Stenvik et al., 2008). Plants were grown until they had at least 15 positions on the inflorescence. A minimum of 15 petals per position were measured. pBS measurements were performed on WT, *rbohd rbohf* (SALK_070610 SALK_059888) and *hae hsl2* (SALK_021905 SALK_030520) plants.

Statistical methods

Two tailed students t-test ($p < 0.05$) was used to identify significant differences in the fluorescent GUS assay by comparing treated samples to untreated samples of the same plant line. Statistical analysis of the RT-qPCR results was performed on all replicas using ONE-WAY or TWO-WAY ANOVA (as stated in the figure text) and post-hoc Tukey's test ($p < 0.05$). Two tailed students t-test with ($p < 0.05$) was used in the petal break strength (pBS) measurements to identify significant differences from WT at a given position on the inflorescence.

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Author contributions

V.O. generated Arabidopsis lines and constructs, tested IDA expression to biotic and abiotic stress, performed gene expression studies, phenotypic analysis of mutants and ROS measurements. V.O. and M.B. performed Ca^{2+} measurements S.G-T. performed the callose deposition assays. V.O., S. G-T., M.B., R.Simon and M.A.B designed experiments, analyzed data, and drafted the manuscript. V.O. and M.A.B wrote the paper with input from all authors.

Competing interests

The authors have no competing interests.

Materials and correspondence

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Supplementary Tables

Supplementary Table 1:

Peptide sequences

Peptide	Amino acid sequence
mIDA	PIPPSA**o**SKRHN
flg22	QRLSTGSRINSAKDDAAGLQIA
IDA$^{\Delta N69}$	PIPPSA**o**SKRH

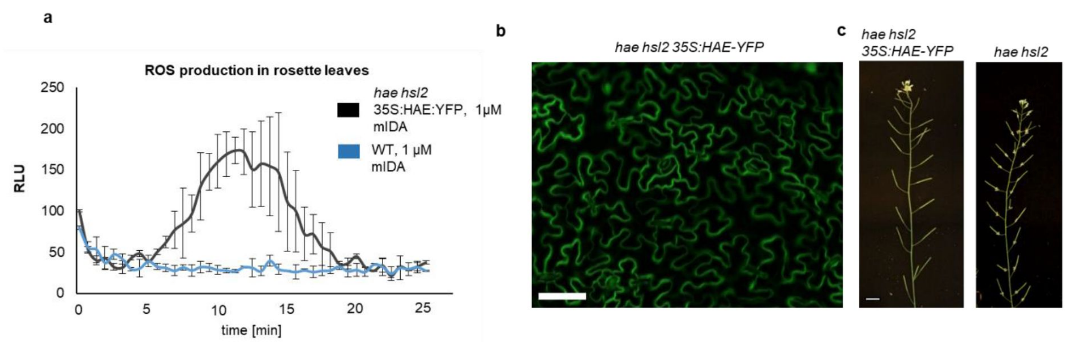
(**o** = hydroxyproline)

Supplementary Table 2:

Primers sequences and function

Function	Primer name	Primer sequence 5'-3'
Cloning promoter FLS2	PromFLS2 -988bp Attb1	5'GGGGACAAGTTTGTACAAAAAAGCAGGCTTA GAAGTTGTGAATTGTGAT'3
Cloning promoter FLS2	PromFLS2 -988bp Attb2	5'GGGGACCACTTTGTACAAGAAAGCTGGGTA GGTTTAGACTTTAGAAGA'3
Genotype hsl2	Hsl2 LP	5'CGTCTTGAGCTAGCCAACAAC'3
Genotype hsl2	Hsl2 RP	5'GTCCAATCAAGTGGAGAAACG'3
Genotype hae	Haesa LP	5'CACCTTCCTTCTCTCCATTCC'3
Genotype hae	Haesa RP	5'GTTCGAGAAGTGACAAGCGAG'3
Genotype SALK-lines	Lbb1-	5'GCGTGGACCGCTTGCTGCAACT'3
Genotype rbohd SALK_070610C	LP_SALK_070610C	5'TTTCAACGCCTTTTGGTACAC'3
Genotype rbohd SALK_070610C	RP_SALK_070610C	5'GTTACCTATTCTTTTGCCGGG'3
Genotype rbohf SALK_059888	LP_SALK_059888	5'CAAAGAGCTCTTCGTGGTTTG'3
Genotype rbohf SALK_059888	RP_SALK_059888	5'TCTCTATTGTATCTTGTGTCACCG'3
Genotype fls2 SALK_062054	FLS2_SALK_062054 Fw	5'GGTTCGATTCCTTCTGGAATC'3
Genotype fls2 SALK_062054	FLS2_SALK_062054 Rv	5'CCTGAGTTTTTGAAGCTTCCC'3
qPCR	33.FRK1 Fw	5'AACTTAGGAGACTATTTGGCAGGTAA'3
qPCR	33.FRK1 Rv	5'TGCATCTAATGATATCTTCAACCTCT'3
qPCR	63.PEP3 Fw	5'GCGAGGAAGATGAGAGTATCG'3
qPCR	63.PEP3 Rv	5'TCAATGGTCATGCCATCTTCT'3
qPCR	91.MYB51 Fw	5'GGCCAATTATCTTAGACCTGACA'3
qPCR	91.MYB51 Rv	5'CCACGAGCTATAGCAGACCATT'3
qPCR reference gene	act2int2 sense	5'CCCTGAGGAGCACCCAGTTCTACTC'3
qPCR reference gene	act2int2 antisense	5'CCGCAAGATCAAGACGAAGGATAGC'3

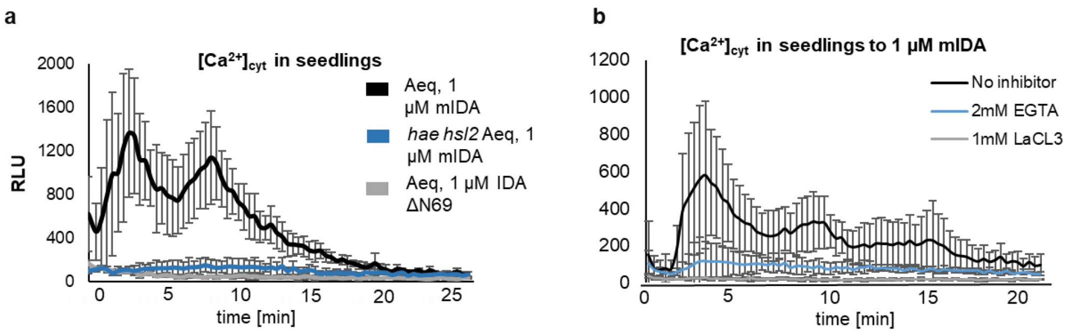
Supplementary figures - A dual function of the IDA peptide in regulating cell separation and modulating plant immunity at the molecular level



Sup Fig. 1:

IDA induces a production of ROS in Arabidopsis

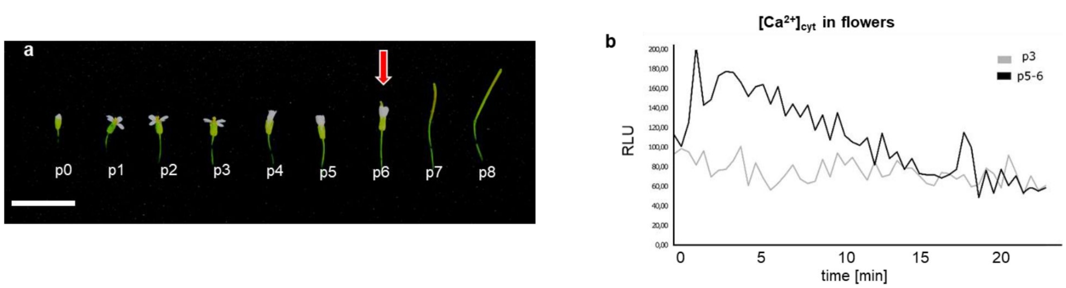
a, ROS production detected by the luminol-based assay was monitored over time as RLU. ROS production from *hae hsl2* leaf disks expressing *35S:HAE-YFP* in response to 1 µM mIDA (black) and in WT control (blue). **b,** Fluorescent image of *hae hsl2* rosette leaves expressing *35S:HAE:YFP*. Signal is detected in the plasma membrane of cells in the epidermal layer. Representative picture. Scale bar = 50 µm. **c,***35S:HAE-YFP* complements the *hae hsl2* mutant abscission phenotype of observed. Scale = 1 cm.



Sup Fig. 2 :

The IDA triggered increase in $[Ca2+]_{cyt}$ is receptor dependent and is abolished in the presence of Ca^{2+} inhibitors

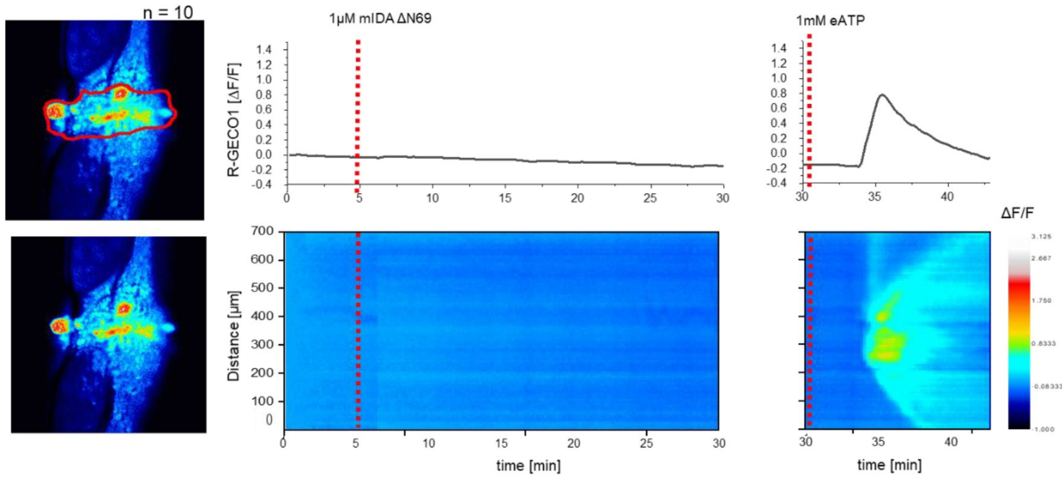
a, Increase in $[Ca^{2+}]_{cyt}$ in seedlings expressing the cytosolic localized *Aequorin*-based luminescence Ca^{2+} sensor (*Aeq*) measured in relative light units (RLU) treated with 1 μM mIDA (black). No response is observed in *Aeq* seedlings treated with 1 μM IDAΔN69 (gray) or in *hae hsl2 Aeq* seedlings treated with 1 μM mIDA (blue). **b,** Increase in $[Ca^{2+}]_{cyt}$ in seedlings expressing the cytosolic localized *Aequorin*-based luminescence Ca^{2+} sensor (*Aeq*) measured in relative light units (RLU) treated with 1 μM mIDA. No response is observed in *Aeq* seedlings treated with 1 mM EGTA (blue) or 1mM $LaCl_3$ (grey). Curves represent average of 3 independent experiments with 4-6 replicates in each experiment.



Sup Fig. 3:

mIDA induces a Ca^{2+} response in flowers

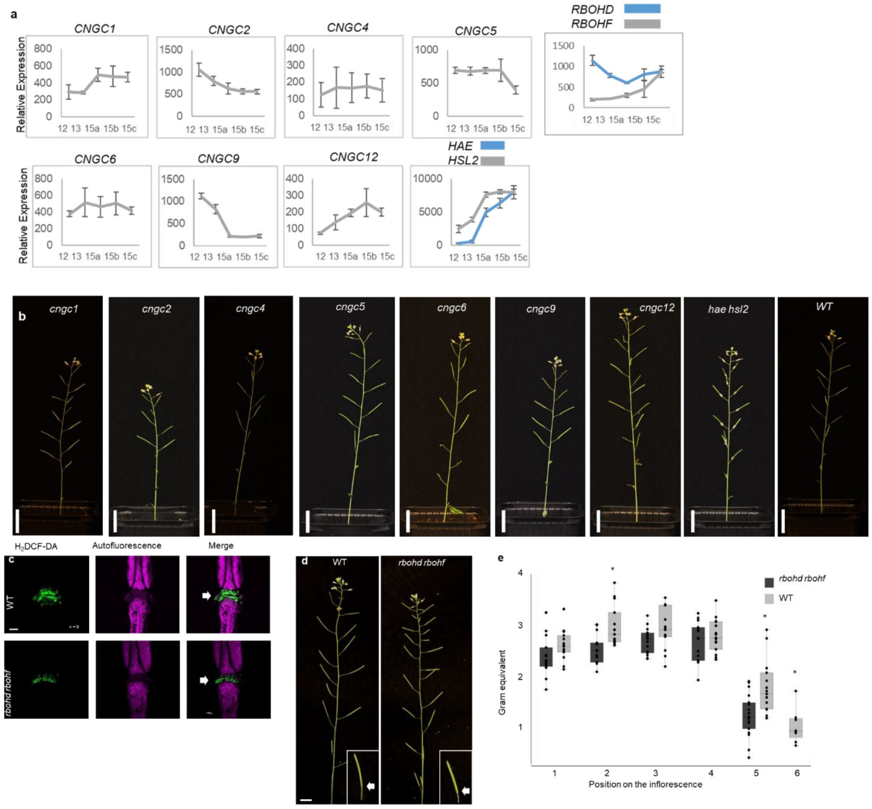
a, Arabidopsis flowers and siliques at different developmental stages. Flowers along the main inflorescence are counted from the first flower with visible white petals at the top of the inflorescence and is defined as p 1. Subsequently older flowers along the inflorescence (p0-p8). Cell wall remodeling in AZ cells, accompanied with a reduction in petal breakstrength (the force required to remove a petal from the receptacle) is initiated at p6 (red arrow). By p7 AZ cells have undergone cell separation and the floral organs have abscised. p=position **b,** Increase in $[Ca^{2+}]_{cyt}$ in flowers expressing the cytosolic localized *Aequorin*-based luminescence Ca^{2+} sensor (*Aeq*) measured in relative light units (RLU) treated with 1 μM mIDA. A $[Ca^{2+}]_{cyt}$ increase is observed in flowers where there is an initial weakening of the cell walls (p5-6) (black). No increase in luminescence was observed in *Aeq* expressing flowers of an earlier developmental stage (p3) (gray). Curves show representative measurements. Three independent experiments were performed with 4-6 replicates in each experiment.



Sup Fig. 4:

mIDAΔN69 does not induce a $[Ca^{2+}]_{cyt}$ release in Arabidopsis abscission zones (AZ)

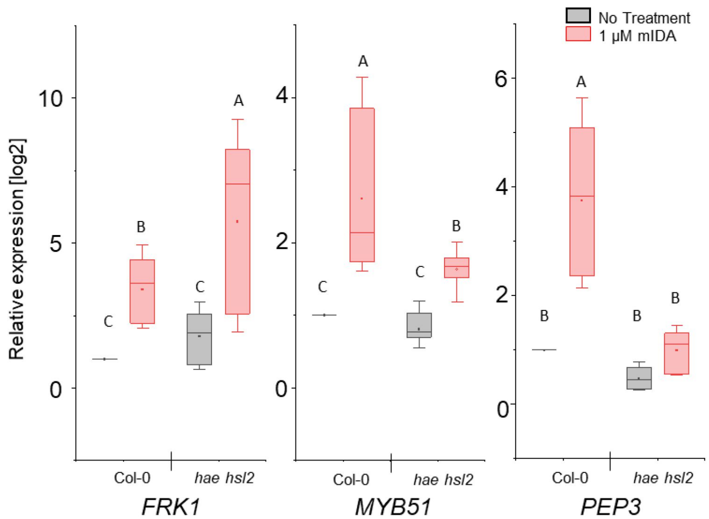
Normalized R-GECO1 fluorescence intensities (ΔF/F) were measured from regions of interest (ROI) (outlined in red) in the floral AZs. Shown are cytosolic calcium concentration ($[Ca^{2+}]_{cyt}$) dynamics in the ROI in response to application of 1 μM mIDAΔN69 or eATP over time. Red lines at 5 minutes (min) indicates application of mIDAΔN69 peptide or application of eATP at 30 min (see also Movie 4). Representative response from 10 flowers. The increase in $[Ca^{2+}]_{cyt}$ response to eATP propagates through the AZ as a single wave seen as normalized R-GECO1 fluorescence intensities (ΔF/F) shown as a heat map.



Sup Fig. 5:

Investigation of CNGCs and RBOHs in the abscission process

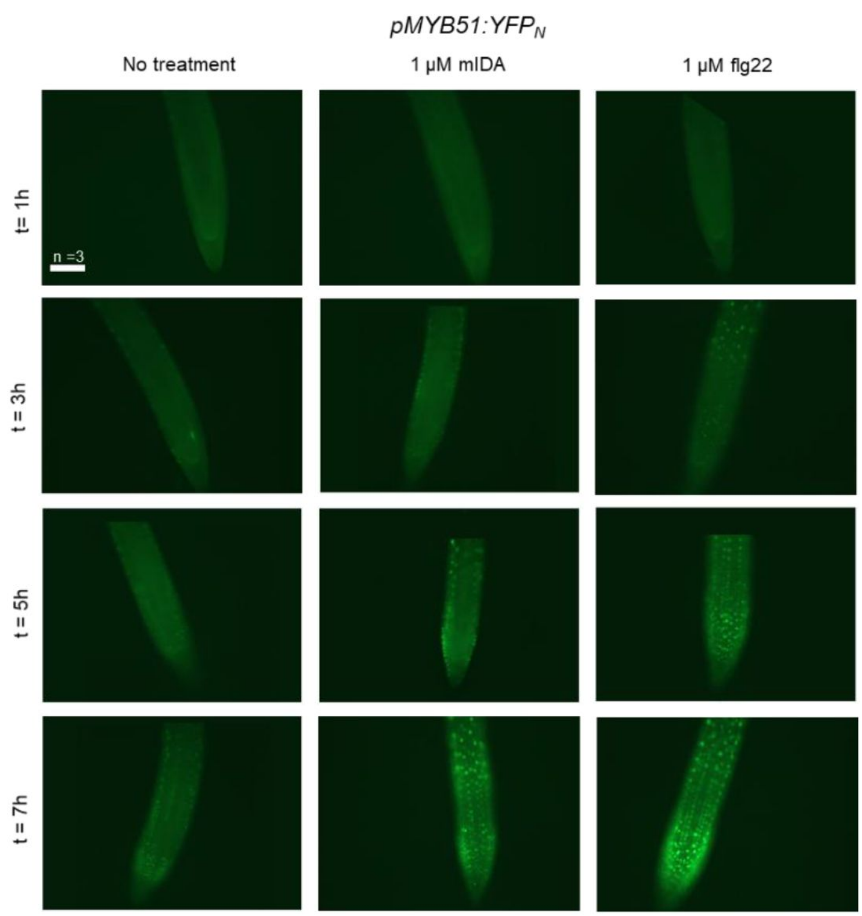
a, Relative expression of *CNGC1*, *CNGC2*, *CNGC4*, *CNGC5*, *CNGC6*, *CNGC9*, *CNGC12, RBOHD* and *RBOHF* in the floral organs during the onset of abscission (Cai and lashbrook 2008). X-axis represents the developmental stages of abscission where position 12 represent flowers where the stamen and petals are of the same length, position 13 represent flowers at anthesis, position 15 represent flowers with stigma extended above the anthers. Floral organ abscission occurs at position 15. Relative expression of *HAE* and *HSL2* added for comparison. Nomenclature for the abscission process and relative expression values taken from (Cai and Lashbrook 2008). **b,** No floral organ abscission phenotype is observed in the single *cngc* mutant plants compared to the *hae hsl2* mutant which retains the floral organ indefinitely. Scale bar = 2cm, 5 weeks old plants. **c,** ROS production in the AZ (white arrow head) detected by using the fluorescent dye H_2DCF-DA in WT and *rbohd rbohf* flowers at position 6 (See Supplementary Fig. 5a for positions). Scale bar = 100 µm, maximum intensity projections of z-stacks. Representative pictures from 9 flowers. **d,** Representative pictures of WT and *rbohd rbohf* inflorescences, scale bar for inflorescences = 1 cm. **e,** Petal break strength (pBS) measurements of WT and the *rbohd rbohf* mutant at position 1-6 along the inflorescence. Force required to remove petals from the receptacle represented in gram equivalent. * = significantly different from WT at the given position ($p < 0.05$, student t-test, two tailed).



Sup Fig 6:

mIDA induced expression of defense-associated marker genes is only partially dependent on the HAE and HSL2 receptors

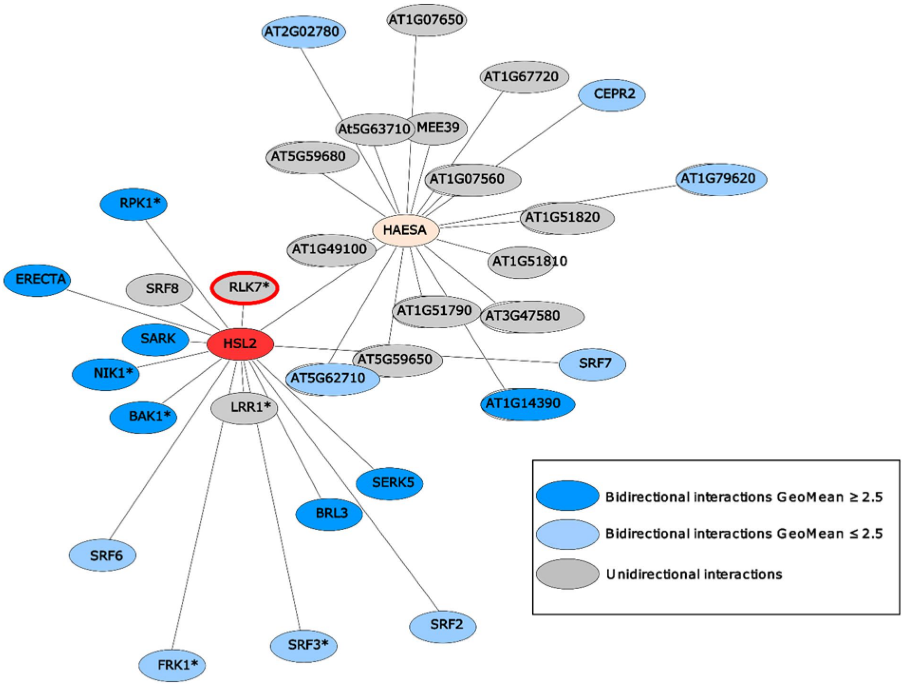
RT-qPCR data showing transcription of *FRK1*, *MYB51*, and *PEP3* in Col-0 WT and *hae hsl2* seedlings exposed to 1 µM mIDA for 1 h compared to untreated seedlings (No treatment). *ACTIN* was used to normalize mRNA levels. Figure represent 3 biological replicates with 4 technical replicates. Statistical analyses was performed comparing samples within same gene, using two-way ANOVA and post-hoc Tukey's test ($p < 0.05$).



Sup Fig. 7:

mIDA and flg22 induced expression of *pMYB51:YFP*

Representative microscopic pictures of 7-days-old *pMYB51:YFPN* roots from seedlings transferred to liquid medium with 1 µM flg22 or 1 µM mIDA for 1, 3, 5 and 7 h using a fluorescent microscope. Control seedlings were transferred to medium with no peptide and imaged in the same time frame. Fluorescent nuclei could be observed in a similar temporal pattern in roots subjected to mIDA and flg22. Scale bar = 100 µm, single plane images using a Zeiss Axioplan2 microscope with an AxioCam HRc, 20x air objective and a YFP filter (Excitation: 500/20, Beamsplitter: FT515, Emission: 535/50), no imaging processing was performed, t = time in h. n = 3, experiment repeated 3 times.



Sup Fig. 8:

HAE (yellow) and HSL2 (red) display very distinct repertoire of immune and growth-related interacting LRR-RKs.

HEA and HSL2 specific interacting partners determined using a sensitized high-throughput extracellular domain interaction assay (Smakowska-Luzan et al,.2018). HSL2 and HAE show distinct interacting LRR-RKs where only HSL2 was found to interact with RLK7 (red halo). * indicates receptors known to play a function in biotic and/or abiotic responses in planta based on previous published work, dark blue nodes refer to bidirectional interactions with GeoMean ≥2.5, light blue nodes-refer to bidirectional interactions with GeoMean ≤2.5, and grey nodes refer to unidirectional interactions.

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Reviewer #1 (Public Review):

The paper titled 'A dual function of the IDA peptide in regulating cell separation and modulating plant immunity at the molecular level' by Olsson Lalun et al., 2023 aims to understand how IDA-HAE/HSL2 signalling modulates immunity, a pathway that has previously been implicated in development. This is a timely question to address as conflicting reports exist within the field. IDL6/7 have previously been shown to negatively regulate immune signalling, disease resistance and stress responses in leaf tissue, however IDA has been shown to positively regulate immunity through the shedding of infected tissues. Moreover, recently the related receptor NUT/HSL3 has been shown to positively regulate immune signalling and disease resistance. This work has the potential to bring clarity to this field, however the manuscript requires some additional work to address these questions. This is especially the case as it contracts some previous work with IDL peptides which are perceived by the same receptor complexes.

Can IDA induce pathogen resistance? Does the infiltration of IDA into leaf tissue enhance or reduce pathogen growth? Previously it has been shown that IDL6 makes plants more susceptible. Is this also true for IDA? Currently cytoplasmic calcium influx and apoplastic ROS as overinterpreted as immune responses - these can also be induced by many developmental cue e.g. CLE40 induced calcium transients. Whilst gene expression is more specific is also true that treatment with synthetic peptides, which are recognised by LRR-RKs, can induce immune gene expression, especially in the short term, even when that is not there in vivo function e.g. doi.org/10.15252/embj.2019103894.

This paper shows that receptors other than hae/hsl2 are genetically required to induce defense gene expression, it would have been interesting to see what phenotype would be associated with higher order mutants of closely related haesa/haesa-like receptors. Indeed recently HSL1 has been shown to function as a receptor for IDA/IDL peptides. Could the triple mutant suppress all response? Could the different receptors have distinct outputs? For example for FRK1 gene expression the hae hsl2 mutant has an enhanced response. Could defence gene expression be primarily mediated by HSL1 with subfunctionalisation within this clade?

One striking finding of the study is the strong additive interaction between IDA and flg22 treatment on gene expression. Do the authors also see this for co-treatment of different peptides with flg22, or is this unique function of IDA? Is this receptor dependent (HAE/HSL1/HSL2)?

It is interesting how tissue specific calcium responses are in response to IDA and flg22, suggesting the cellular distribution of their cognate receptors. However, one striking observation made by the authors as well, is that the expression of promoter seems to be broader than the calcium response. Indicating that additional factors are required for the observed calcium response. Could diffusion of the peptide be a contributing factor, or are only some cells competent to induce a calcium response?
It is interesting that the authors look for floral abscission phenotypes in cngc and rbohd/f mutants to conclude for genetic requirement of these in floral abscission. Do the authors have a hypothesis for why they failed to see a phenotype for the rbohd/f mutant as was published previously? Do you think there might be additional players redundantly mediating these processes?

Can you observe callose deposition in the cotyledons of the 35S::HAE line? Are the receptors expressed in native cotyledons? This is the only phenotype tested in the cotyledons.

Are flg22-induced calcium responses affected in hae hsl2?

Reviewer #2 (Public Review):

Lalun and co-authors investigate the signalling outputs triggered by the perception of IDA, a plant peptide regulating organs abscission. The authors observed that IDA perception leads to a transient influx of Ca2+, to the production of reactive oxygen species in the apoplast, and to an increase accumulation of transcripts which are also responsive to an immunogenic epitope of bacterial flagellin, flg22. The authors show that IDA is transcriptionally upregulated in response to several biotic and abiotic stimuli. Finally, based on the similarities in the molecular responses triggered by IDA and elicitors (such as flg22) the authors proposed that IDA has a dual function in modulating abscission and immunity. The manuscript is rather descriptive and provide little information regarding IDA signalling per se. A potential functional link between IDA signalling and immune signalling remains speculative.

Reviewer #3 (Public Review):

Previously, it has been shown the essential role of IDA peptide and HAESA receptor families in driving various cell separation processes such as abscission of flowers as a natural developmental process, of leaves as a defense mechanism when plants are under pathogenic attack or at the lateral root emergence and root tip cell sloughing. In this work, Olsson et al. show for the first time the possible role of IDA peptide in triggering plant innate immunity after the cell separation process occurred. Such an event has been previously proposed to take place in order to seal open remaining tissue after cell separation to avoid creating an entry point for opportunistic pathogens. The elegant experiments in this work demonstrate that IDA peptide is triggering the defense-associated marker genes together with immune specific responses including release of ROS and intracellular CA2+. Thus, the work highlights an intriguing direct link between endogenous cell wall remodeling and plant immunity. Moreover, the upregulation of IDA in response to abiotic and especially biotic stimuli are providing a valuable indication for potential involvement of HAE/IDA signalling in other processes than plant development.

Strengths:
The various methods and different approaches chosen by the authors consolidates the additional new role for a hormone-peptide such as IDA. The involvement of IDA in triggering of the immunity complex process represents a further step in understanding what happens after cell separation occurs. The Ca2+ and ROS imaging and measurements together with using the haehsl2 and haehsl2 p35S::HAE-YFP genotypes provide a robust quantification of defense responses activation. While Ca2+ and ROS can be detected after applying the IDA treatment after the occurrence of cell separation it is adequately shown that the enzymes responsible for ROS production, RBOHD and RBOHF, are not implicated in the floral abscission.
Furthermore, IDA production is triggered by biotic and abiotic factors such as flg22, a bacterial elicitor, fungi, mannitol or salt, while the mature IDA is activating the production of FRK1, MYB51 and PEP3, genes known for being part of plant defense process.

Weaknesses:
Even though there is shown a clear involvement of IDA in activating the after-cell separation immune system, the use of p35S:HAE-YFP line represent a weak point in the scientific demonstration. The mentioned line is driving the HAE receptor by a constitutive promoter, capable of loading the plant with HAE protein without discriminating on a specific tissue. Since it is known that IDA family consist of more members distributed in various tissues, it is very difficult to fully differentiate the effects of HAE present ubiquitously.
The co-localization of HAE/HSL2 and FLS2 receptors is a valuable point to address since in the present work, the marker lines presented do not get activated in the same cell types of the root tissues which renders the idea of nanodomains co-localization (as hypothetically written in the discussion) rather unlikely.